

4 Forces, density and pressure

AS · 2026 examination · Concept guide and original exam-style practice

Understand the concept

Force diagrams must show forces on one chosen body. Resolve vectors before finding the resultant or equilibrium. The moment about a pivot is force times perpendicular lever arm. Density is mass/volume and hydrostatic pressure change $\Delta p = \rho g \Delta h$; pressure acts normally to a surface.

Worked example: follow the reasoning

A 90 N load 0.40 m from a hinge gives moment 36 N m. A support 0.60 m from the hinge must exert 60 N in the opposite turning direction.

Quick check · multiple choice

A Pressure increase at depth h in a liquid is proportional to what? [1]

- A $1/h$
- B h^2 only
- C Surface area
- D ρgh

Write your choice: _____

Structured practice

Answer in the space provided. Show working and units where relevant.

1 A water tank is filled to depth 1.8 m. Find the pressure increase from surface to bottom, using $\rho = 1000 \text{ kg m}^{-3}$ and $g = 9.81 \text{ m s}^{-2}$. [2]

Final answer: _____

Answers and explanation

Quick check: D. ρgh

Question 1 [2 marks]

$$\Delta p = \rho gh [1] = 1000 \times 9.81 \times 1.8 = 1.77 \times 10^4 \text{ Pa (3 s.f.) [1]}.$$

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or calculation error; then retry without looking at the solution.

Syllabus: Physics 9702, 2025–2027. <https://www.cambridgeinternational.org/Images/664565-2025-2027-syllabus.pdf>

Original FagusED teaching material. Question layout and mark presentation informed by the May/June 2026 papers supplied by the user. The practice question is newly written and is not a Cambridge past-paper question.